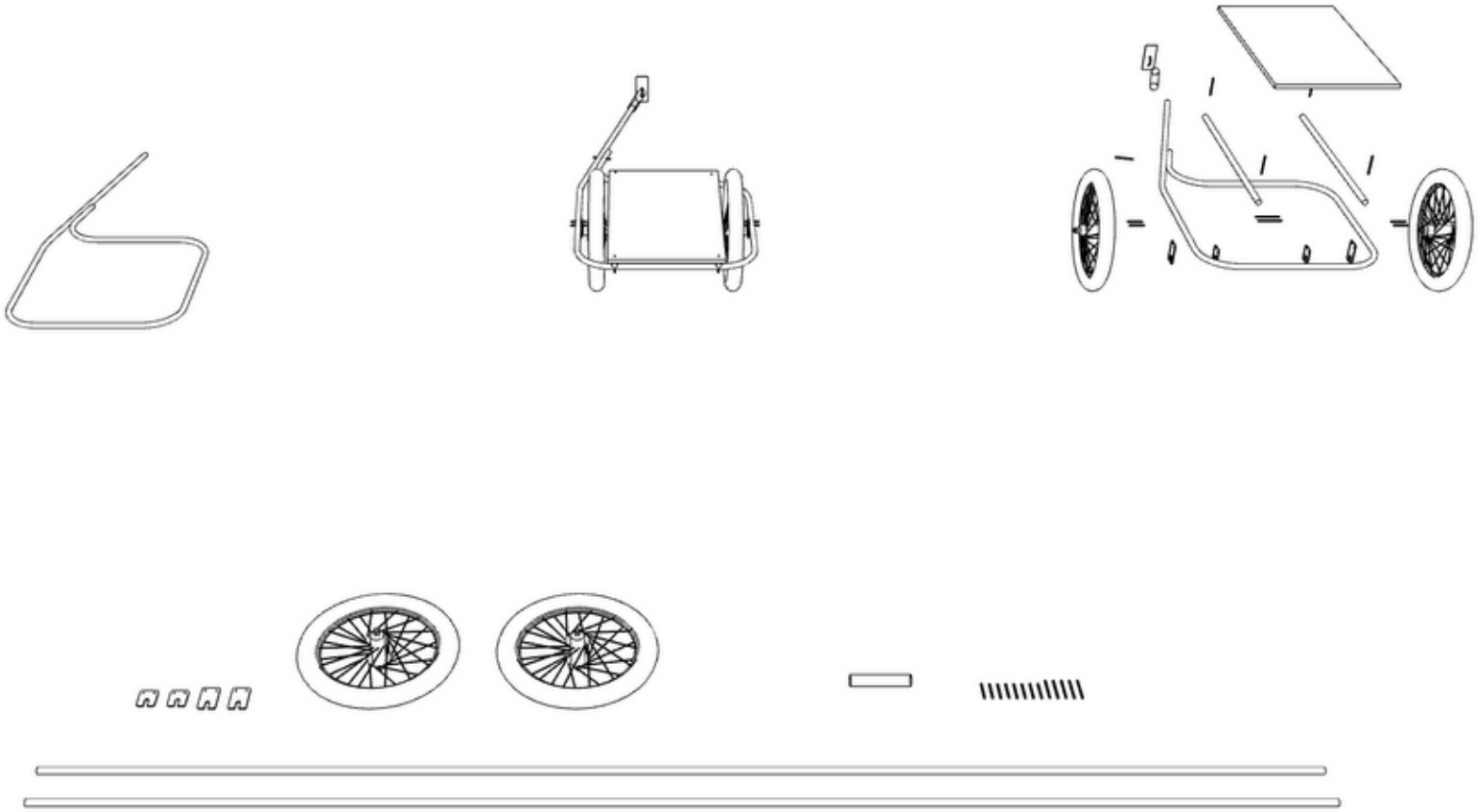
Bike Trailer v1.0 (Edge Collective)

1/2 in EMT conduit frame | recycled 16 in wheels | laser-cut dropouts | vinyl-tube hitch

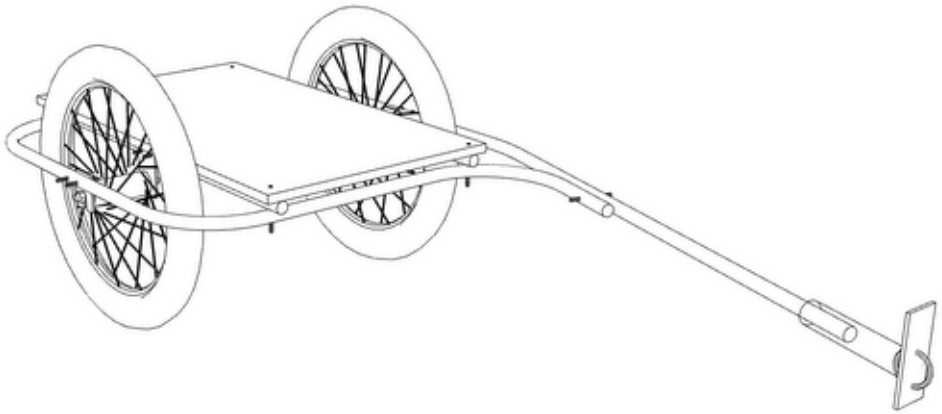


1 Materials > 2 Bent main frame > 3 Assembled trailer > 4 Exploded

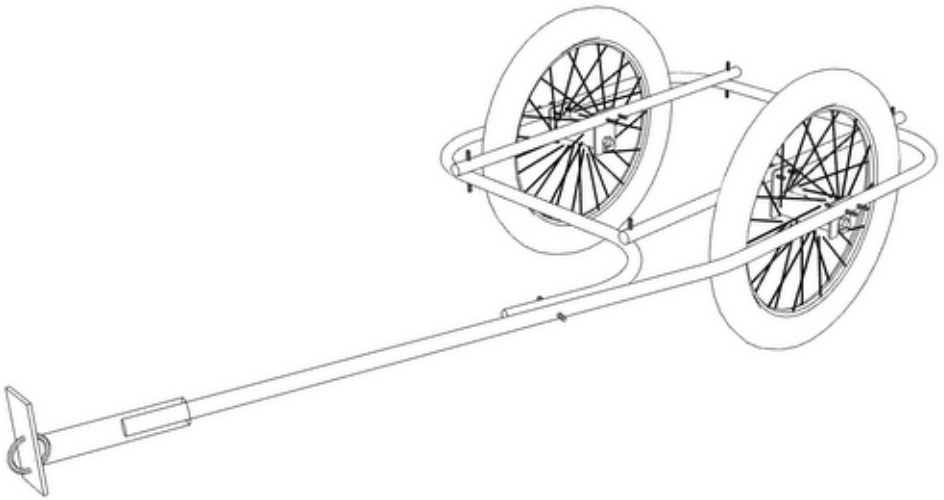
- Main frame: one 10 ft length of 1/2 in EMT, bent into the tongue and a loop around both wheels (about 23 3/4 x 23 3/4 in).
- Inner bars: two 25 in pieces from a second 10 ft length, bolted on top of the loop.
- Each wheel hangs between a short dropout (outer bar) and a long dropout (inner bar, one pipe higher).
- Hitch: 6 in of 7/8 in braided vinyl tube over the tongue tip, to a Burley-style coupler on the bike axle.
- Frame sizes are estimates from the project photos; dropouts follow the project's SCAD files.

The assembled trailer

With the plywood deck, and the bare frame



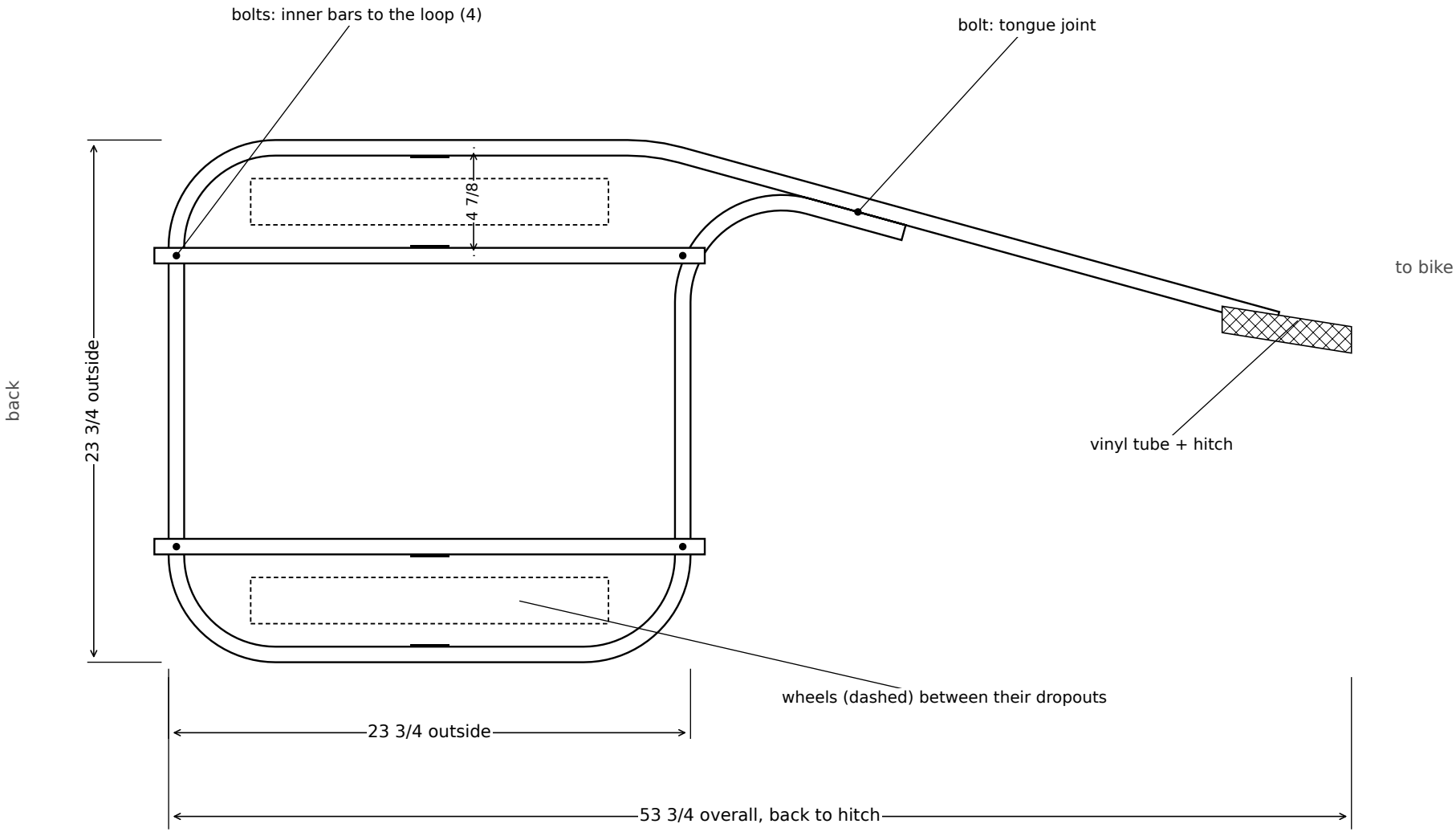
Assembled, with the deck



Frame without the deck

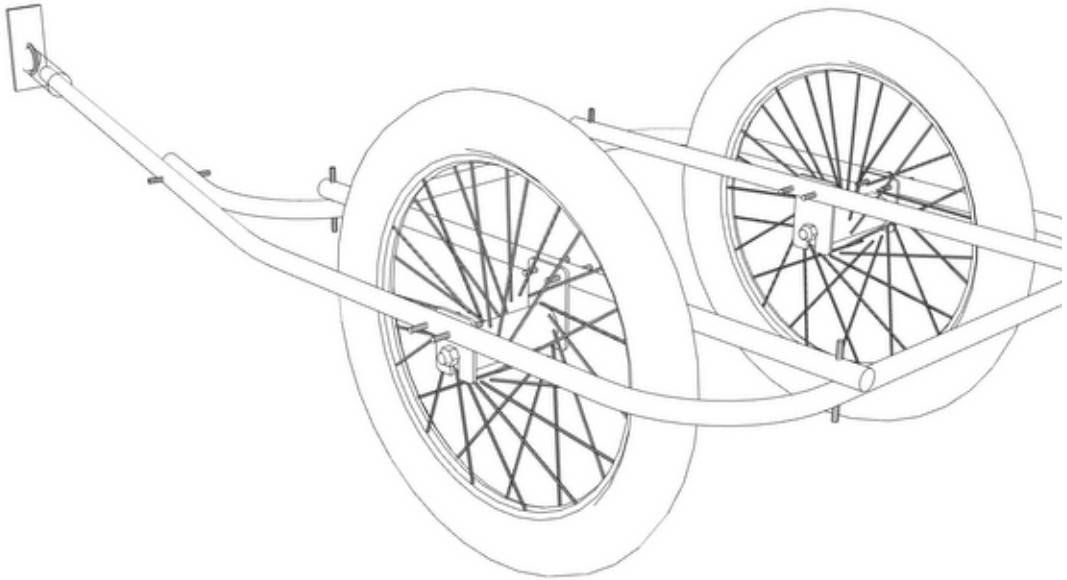
Frame, from above (1:7)

Main frame (one 10 ft conduit), inner bars, wheels dashed, dropouts

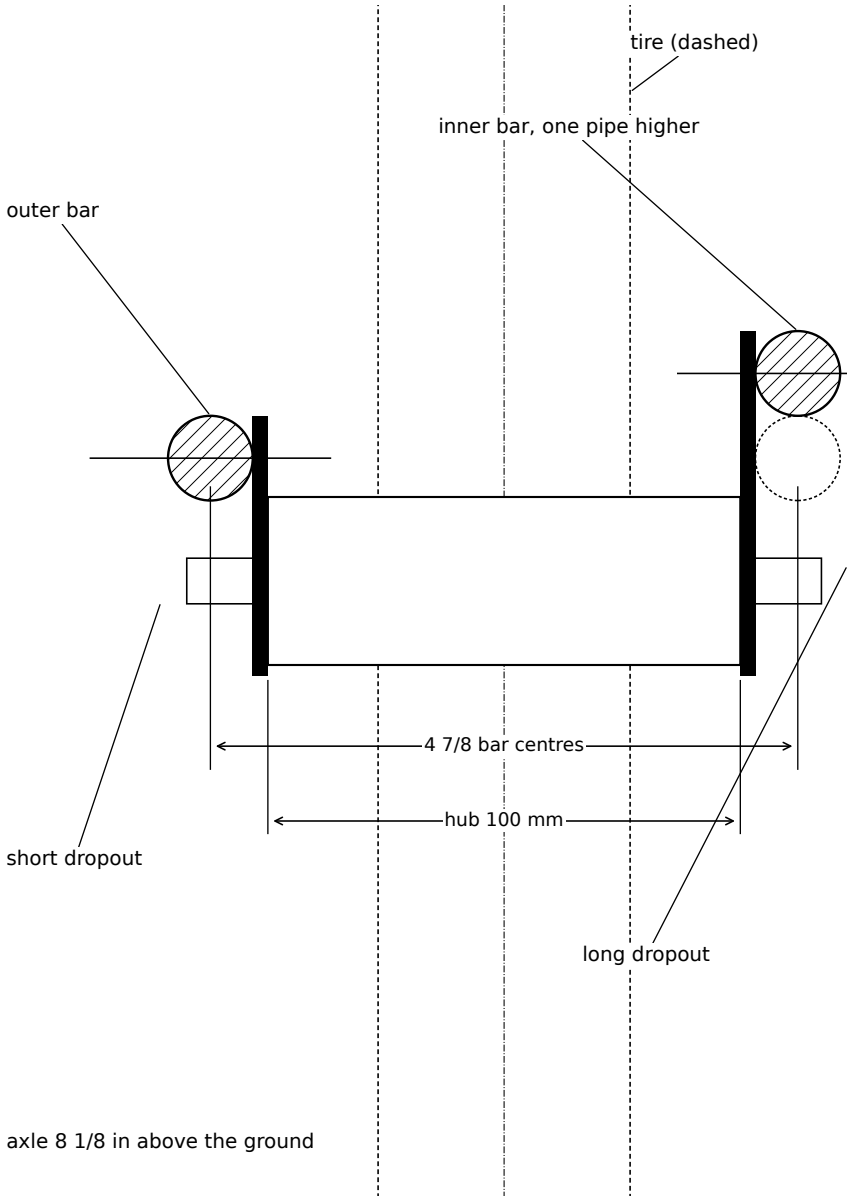


The wheel mount

Short dropout on the outer bar, long dropout on the inner bar, axle level

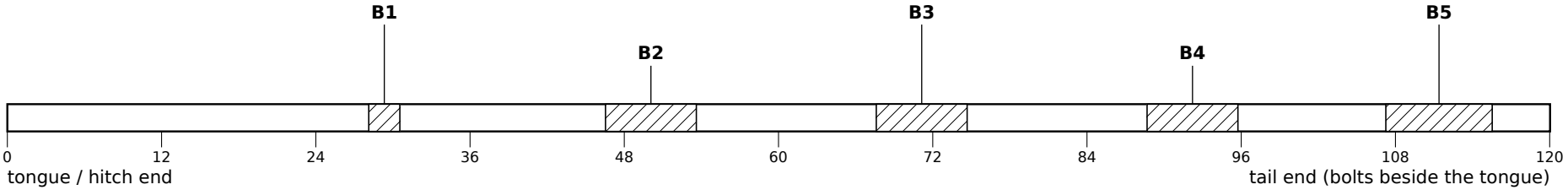


Section through the axle, looking forward (1:1.6)



Bending the main frame (one 10 ft conduit)

Marks measured along the pipe from the tongue (hitch) end, centreline; all bends in one plane



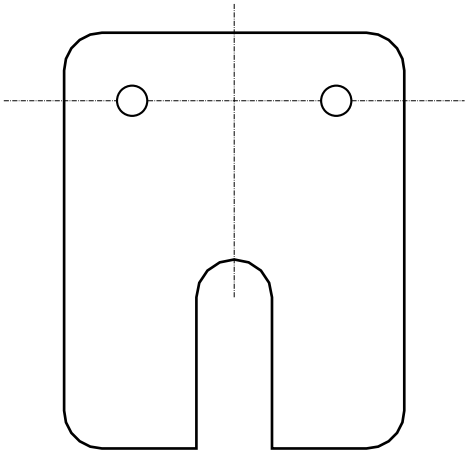
	Start (in)	End (in)	Angle	Turn*	What it makes
B1	28 1/8	30 1/2	15 deg	left	tongue into the left outer bar
B2	46 1/2	53 5/8	90 deg	left	back left corner
B3	67 5/8	74 5/8	90 deg	left	back right corner
B4	88 5/8	95 3/4	90 deg	left	front right corner
B5	107 1/4	115 1/2	105 deg	right	front cross member into the tongue tail

* Turn direction as you walk the pipe from the tongue end, frame lying flat. Bend radius about 4 1/2 in (1/2 in EMT hand bender); B1 is a gentle bend. Allow for your bender's take-up when marking.

Dropouts, full size (1:1)

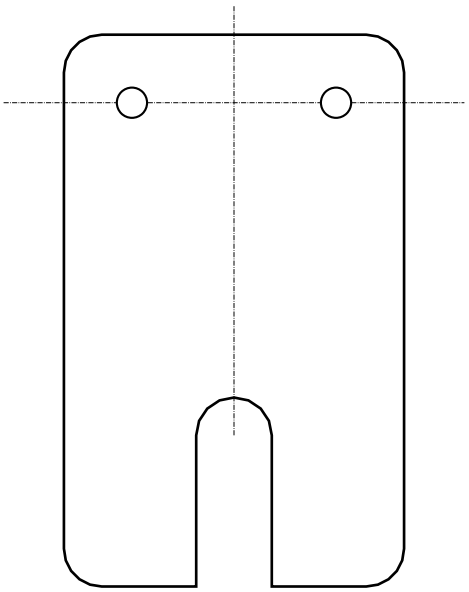
Print at 100% (no scaling). 3.2 mm (0.125 in) stainless; check the squares below

Short dropout (outer bar), make 2

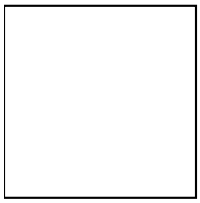


45 x 55 mm (1 3/4 x 2 3/16 in)
2 holes 4 mm (#10 bolts), 27 mm apart
axle slot 10 mm wide

Long dropout (inner bar), make 2

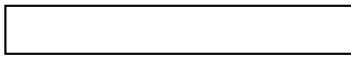


45 x 73 mm (1 3/4 x 2 7/8 in)
2 holes 4 mm (#10 bolts), 27 mm apart
axle slot 10 mm wide



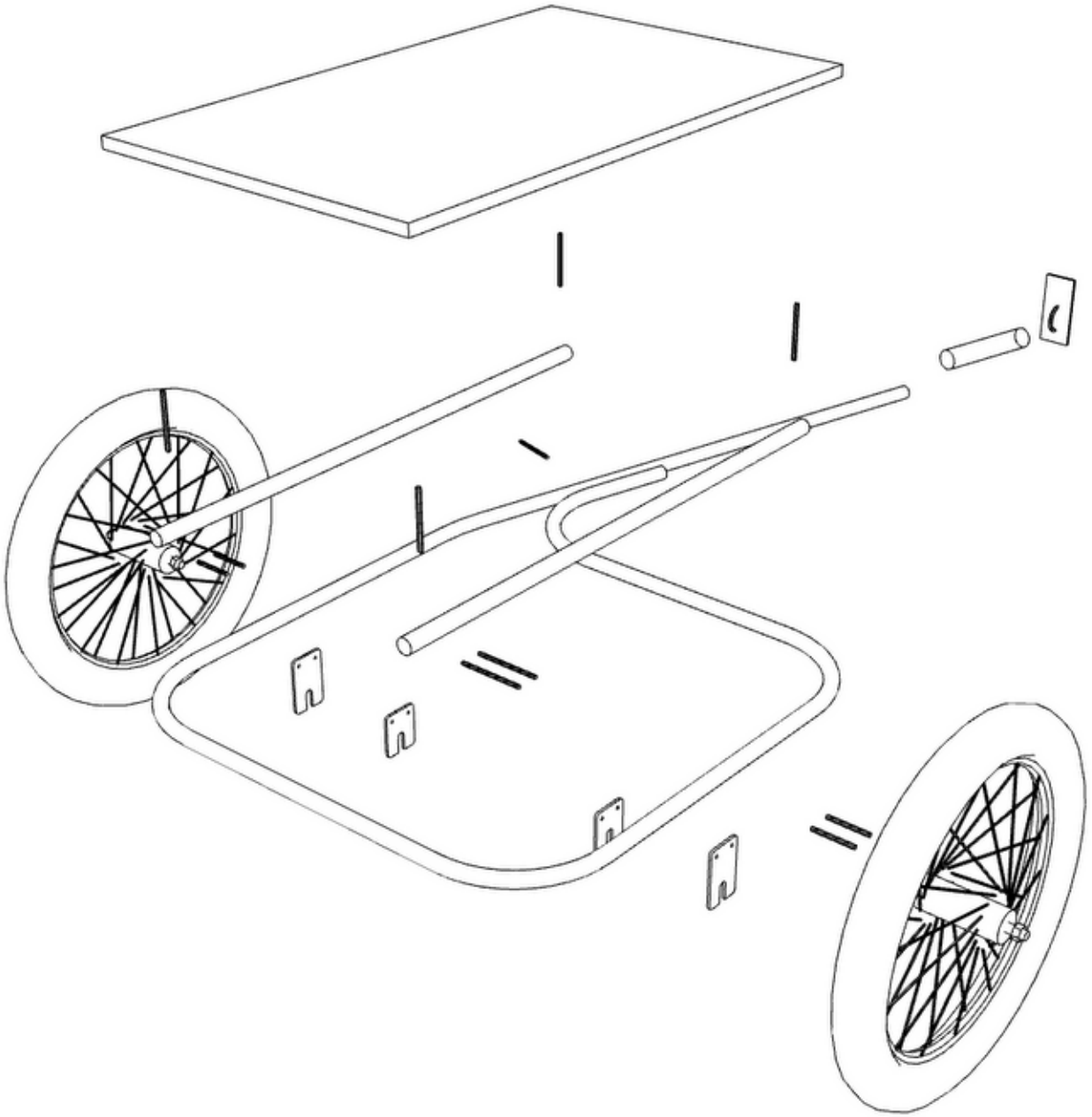
1 in square

50 mm



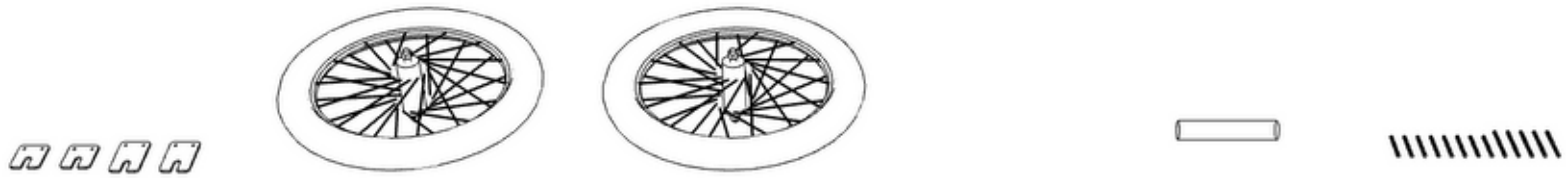
Every part, exploded

Main frame, inner bars, bolts, dropouts, wheels, hitch and deck



Parts list

From the project page; quantities match the model



Everything laid out before building

Part	Qty	Notes
1/2 in EMT conduit, 10 ft	2	one for the main frame (all 120 in); inner bars 2 x 25 in from the other
Dropout, short (outer bar)	2	laser cut, 0.125 in stainless (e.g. SendCutSend)
Dropout, long (inner bar)	2	same, one pipe diameter (18 mm) longer
Bolt #10-24 x 2 in + nut	8	dropouts to bars, 2 each
Bolt #10-24 x 2 1/2 in + nut	5	inner bars to the loop (4), tongue joint (1)
16 in wheel	2	recycled kids' bike wheels, 100 mm hubs
Braided vinyl tube 7/8 in	6 in	over the tongue tip
Bike trailer hitch coupler	1	Burley-style, on the bike's rear axle
Plywood deck 1/2 in, 15 x 27 in	1	optional
Tools		1/2 in EMT bender, drill + 11/64 in bit, screwdriver, pliers